

연성범 (Seongbeom Yeon)

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Education

Sep. 2021 – Present	Integrated M.S. and Ph.D., Materials Science and Engineering KAIST, Daejeon, Korea (Advisor: Professor Himchan Cho)
Sep. 2014 – Feb. 2021 (Military Service: Feb. 2016 – Nov. 2017)	Integrated B.S. and M.S., Chemistry Chungbuk National University, Cheongju, Korea

Thesis title: (An) Experimental and Theoretical Study for Synthesis of new n-type Zintl Phase Compounds : The $\text{Ca}_{5-x-y}\text{Yb}_x\text{RE}_y\text{Al}_2\text{Sb}_6$ (RE =Pr, Nd, Sm, Gd, Eu; $0 \leq x \leq 4.82(4)$; $0 \leq y \leq 0.44(1)$), Advisor: Prof. Tae-Soo You

Research Highlightes

† (equal first author)

- Changhyun Joo†, **Seongbeom Yeon**†, Jordi Llusar, Shehla Gul, Chanhoo Park, Junhui Lee, Jaeyeong Ha, Jaedong Jang, In Hye Kwak, Hyung Joong Yun, Sang Kyu Kim, Yong-Hoon Cho, Ivan Infante*, Himchan Cho*, “Overcoming the Luminescence Efficiency Limitations of InP Magic-Sized Clusters”
Journal of the American Chemical Society **2025**, 52, 48046–48059 (SCI, IF = 16.6)
- Seongbeom Yeon**, Yoseph Kim, Abdessamad El Adel, Jaeyeong Ha, Seongkyu Maeng, Youngjo Kim, Ivan Infante*, Himchan Cho*, “Olefin Ligand Metathesis for Colloidal Emissive Nanocrystals with Enhanced Stability and Photosensitivity”
Angewandte Chemie International Edition **2025**, e202514802 (SCI, IF = 16.9)
(Selected as a **Hot paper**)
- Jaehwan Lee†, **Seongbeom Yeon**†, Bonwoo Koo, Bokyeong Sohn, Sun Jae Park, Cheoljae Kim, Himchan Cho*, “Photocleavable Ligand-Induced Direct Photolithography of InP-Based Quantum Dot”
ACS Energy Letters **2025**, 10, 94–101 (SCI, IF = 19.5)
(Selected as a **supplementary cover**)

4. Jaeyeong Ha[†], **Seongbeom Yeon[†]**, Jaehwan Lee, Hyungdoh Lee, Himchan Cho*,
“Revealing the Role of Organic Ligands in Deep-Blue-Emitting Colloidal Europium
Bromide Perovskite Nanocrystals”
ACS Nano **2024**, 18, 31891-31902 (SCI, IF = 15.881)
(Selected as a **supplementary cover**)
5. **Seongbeom Yeon**, Seungeun Shin, Hongil Jo, Nguyen Le Thi, Dohyun Moon, Dong-Hyun
Kim, Kang Min Ok, Tae-Soo You*, “p-Type to n-Type Conversion through the “Bypass”
Phase Transition in the Zintl-Phase Thermoelectric Materials”
Chem. Mater. **2021**, 33, 6761–6773 (SCI, IF = 10.508)

Selected Honors & Awards

[†] Joint winners (equally contributed 1st author)

Jun. 2026	GCIM 2026 Best Poster Presentation Award <i>International Conference</i> Materials Research Society of Korea, Jeju, Korea
May. 2026	QD2026 Best Poster Presentation Award[†] <i>International Conference</i> The Korean Information Display Society, Busan, Korea
Jul. 2025	Excellent Paper Presentation Award[†] Materials Research Society of Korea, Gangneung, Korea
Apr. 2024	MRS Spring 2024 Best Poster Award[†] <i>International Conference</i> Materials Research Society (MRS) Spring Meeting & Exhibit, Seattle, Washington, USA
Nov. 2023	Excellent Paper Presentation Award[†] Korean Society of Industrial and Engineering Chemistry, Gwangju, Korea
Nov. 2022	Excellent Paper Presentation Award Korean Society of Industrial and Engineering Chemistry, Daejeon, Korea
Nov. 2021	Best Paper Award Korean Chemical Society, Seoul, Korea
May. 2019	Excellent Oral Presentation Award College of Natural Sciences, Chungbuk National University, Cheongju, Korea

Scholarships

Sep. 2021 – Present	Full Scholarship KAIST, Daejeon, Korea
Sep. 2019 – Aug. 2020	Full Scholarship Chungbuk National University, Cheongju, Korea
Mar. 2016 – Aug. 2019	National Scholarship for Science and Engineering Ministry of Science and ICT, Sejong, Korea
Sep. 2015	Scholarship for Knowledge Mentoring Chungbuk National University, Cheongju, Korea
Mar. 2015 – Feb. 2016	Scholarship for Outstanding Student Chungbuk National University, Cheongju, Korea
Mar. 2014 – Feb. 2015	National Scholarship Korea Student Aid Foundation, Seoul, Korea

Professional Activities

Sep. 2022	<i>Organization Staff</i> The Global Networking Workshop for New Researchers of Nanoelectronic Materials Online Meeting
Fall 2021	<i>Teaching Assistant</i> General Chemistry I KAIST, Daejeon, Korea
Nov. 2020	<i>Organization Staff</i> The 2020 Central Region Inorganic Chemistry Symposium Chungbuk National University, Cheongju, Korea
Fall 2020	<i>Teaching Assistant</i> Inorganic Chemistry I Chungbuk National University, Cheongju, Korea
Spring 2020	<i>Teaching Assistant</i> General Chemistry I Chungbuk National University, Cheongju, Korea
Jan. 2020	<i>Organization Staff</i> The 2020 Winter Inorganic Solid-State Chemistry Workshop Gwangju-si, Gyeonggi-do, Korea
Fall 2019	<i>Teaching Assistant</i> General Chemistry II Chungbuk National University, Cheongju, Korea
Jan. 2019	<i>Organization Staff</i> The 2019 Winter Inorganic Solid-State Chemistry Workshop Pyeongchang, Korea

Full Publications

† (equal first author)

1. Changhyun Joo†, **Seongbeom Yeon†**, Jordi Llusar, Shehla Gul, Chanho Park, Junhui Lee, Jaeyeong Ha, Jaedong Jang, In Hye Kwak, Hyung Joong Yun, Sang Kyu Kim, Yong-Hoon Cho, Ivan Infante*, Himchan Cho*, “Overcoming the Luminescence Efficiency Limitations of InP Magic-Sized Clusters”
Journal of the American Chemical Society **2025**, 52, 48046–48059 (SCI, IF = 16.6)
2. Jaehwan Lee†, Yonghan Jo†, **Seongbeom Yeon**, Jinha Jang, Jaeyeong Ha, Bokyeong Sohn, Hyunjin Cho, Gui-Min Kim, Doh C. Lee, Chan Beum Park*, and Himchan Cho*, “All-Inorganic InP/ZnSe/ZnS Quantum Dots with Chalcogenidometallate Ligands for Biosolar Catalysis”
Advanced Functional Materials **2025**, e11671 (SCI, IF = 19)
3. **Seongbeom Yeon**, Yoseph Kim, Abdessamad El Adel, Jaeyeong Ha, Seongkyu Maeng, Youngjo Kim, Ivan Infante*, Himchan Cho*, “Olefin Ligand Metathesis for Colloidal Emissive Nanocrystals with Enhanced Stability and Photosensitivity”
Angewandte Chemie International Edition **2025**, e202514802 (SCI, IF = 16.9)
(Selected as a **Hot paper**)
4. Jaehwan Lee†, **Seongbeom Yeon†**, Bonwoo Koo, Bokyeong Sohn, Sun Jae Park, Cheoljae Kim, Himchan Cho*, “Photocleavable Ligand-Induced Direct Photolithography of InP-Based Quantum Dot”
ACS Energy Letters **2025**, 10, 94–101 (SCI, IF = 19.5)
(Selected as a **supplementary cover**)
5. Jaeyeong Ha†, **Seongbeom Yeon†**, Jaehwan Lee, Hyungdoh Lee, Himchan Cho*, “Revealing the Role of Organic Ligands in Deep-Blue-Emitting Colloidal Europium Bromide Perovskite Nanocrystals”
ACS Nano **2024**, 18, 31891–31902 (SCI, IF = 15.881)
(Selected as a **supplementary cover**)
6. Junsu Lee, Naeun Seo, **Seongbeom Yeon**, Tae-Soo You*, “Novel n-type Zintl Phase Thermoelectric Materials: The $\text{Ca}_{5-x}\text{RE}_x\text{Al}_2\text{Sb}_6$ (RE = Pr, Sm)”
Bull. Kor. Chem. Soc. **2023**, 44, 705–712 (SCI)
7. Sujin Kim, Junphil Hwang, Tae-Soo You, **Seongbeom Yeon**, Jungwon Kim, Byung-Kyu Yu, Mi-Kyung Han, Minju Lee, Somnath Acharya, Jiyong Kim, Woochul Kim, and Sung-Jin Kim*, “Enhanced Thermoelectric Performance by Resonant Doping and Embedded

Magnetic Impurity”

Phys. Rev. Applied. **2022**, 19, 014034. (SCI, IF = 4.7)

8. Yeongjin Hong, **Seongbeom Yeon**, Philip Yox, Zhao Yunxiu, Dohyun Moon, KangMin Ok, Dong-Hyun Kim, Kirill Kovnir, Gordon J. Miller, and Tae-Soo You*, “Role of Eu-Doping on the electron Transport Behavior in the Zintl Thermoelectric $\text{Ca}_{5-x-y}\text{Yb}_x\text{Eu}_y\text{Al}_2\text{Sb}_6$ System” **Chem. Mater.** **2022**, 34, 9903–9914. (SCI, IF = 8.6)
(Selected as a **front cover**)
9. Yeongjin Hong[†], **Seongbeom Yeon**[†], Jiwon Jeong, Dohyun Moon, and Tae-Soo You*, “Rare-Earth Metal Doped Zintl Phase Thermoelectric Materials: the $\text{Yb}_{5-x}\text{RE}_x\text{Al}_2\text{Sb}_6$ (RE = Pr, Nd, Sm) System” **Bull. Kor. Chem. Soc.** **2022**, 43, 1191–1199 (SCI, Selected as a **front cover**)
10. Jee Yoon Chung, **Seongbeom Yeon**, Hongsun Ryu, Tae-Soo You*, Joon I. Jang*, Kang Min Ok*, “Nonlinear Optical Properties of A New Polar Bismuth Tellurium Oxide Fluoride $\text{Bi}_3\text{F}(\text{TeO}_3)(\text{TeO}_2\text{F}_2)_3$ ” **J. Alloys Compd.** **2022**, 895, 162603. (SCI, IF = 6.371)
11. **Seongbeom Yeon**, Seungeun Shin, Hongil Jo, Nguyen Le Thi, Dohyun Moon, Dong-Hyun Kim, Kang Min Ok, Tae-Soo You*, “p-Type to n-Type Conversion through the “Bypass” Phase Transition in the Zintl-Phase Thermoelectric Materials” **Chem. Mater.** **2021**, 33, 6761–6773 (SCI, IF = 10.508)
12. Junsu Lee, Yeongjin Hong, Seongbeom Yeon, Dohyun Moon, Tae-Soo You*, “Effect of Cationic and Anionic Doping in the Quinary Zintl Phase Thermoelectric Material $\text{Ca}_{5-x}\text{Yb}_x\text{Al}_{2-y}\text{In}_y\text{Sb}_{6-z}\text{Sn}_z$ System” **Bull. Kor. Chem. Soc.** **2021**, 42, 563–566. (SCI)
13. Junsu Lee, Seongbeom Yeon, Tae-Soo You*, “Development of Thermoelectric Materials Converting Various Thermal Energy into Electricity” **Chemworld** **2020**, 7, 25–34. / Domestic academic journals
14. Jee Yoon Chung, Hongil Jo, Seongbeom Yeon, Hye Ryung Byun, Tae-Soo You, Joon I. Jang, Kang Min Ok*, “ $\text{Bi}_3(\text{SeO}_3)_3(\text{Se}_2\text{O}_5)\text{F}$: A Polar Bismuth Selenite Fluoride with Polyhedra of Highly Distortive Lone Pair Cations and Strong Second-Harmonic Generation Response” **Chem. Mater.** **2020**, 32, 7318–7326. (SCI, IF = 9.811)
(Selected as a **supplementary cover**)

Patents

1. Domestic granted patents (국내 특허 등록)

- 1) 유태수, 연성범, “희토류 금속 양이온, 칼슘 양이온, 및 이터븀 양이온이 포함된 오성분계 진틀상 화합물 및 이의 제조방법 (Rare-earth Metal Cation, Calcium Cation, And Ytterbium Cation Mixed Quinary Zintl Phase Compound And Manufacturing Method Thereof)”, Korea, Application No.: 10-2021-0078003 (2021.06.16), **Grant: 1025235450000 (2023.04.14)**
- 2) 유태수, 연성범, “유로퓸이 도입된 진틀화합물 및 이의 제조방법 (Eu-Doping Zintl Compound And Manufacturing Method Thereof)”, Korea, Application No.: 10-2022-0103603 (2022.08.18), **Grant: 1024922020000 (2023.01.19)**

2. Domestic patent applications (국내 특허 출원)

- 1) 조힘찬, 연성범, “올레핀 복분해 반응을 이용한 무기 나노입자의 개질 방법, 무기 나노입자를 포함하는 감광성 조성물 및 이를 이용한 패턴 형성 방법”, Korea, **Application No.: 10-2025-0026528 (2025.02.28)**
- 2) 조힘찬, 이재환, 연성범, “광절단 가능한 광민감성 리간드 기반의 양자점 패터닝 기술 (Quantum dots containing light-activated ligands, a quantum dot patterning method using the same, and an electroluminescent device)”, Korea, **Application No.: 10-2024-0157970 (2024.11.08)**
- 3) 조힘찬, 연성범, “무기 다중 금속 칼코제나이드 복합체 리간드를 갖는 나노입자, 이를 포함하는 광감성 조성물 및 이를 이용한 패턴 형성 방법 (NANO-PARTICLE HAVING INORGANIC MULTINARY METAL CHALCOGENIDE COMPLEX LIGAND, PHOTSENSITIVE COMPOSITION INCLUDING THE SAME AND METHOD FOR FORMING PATTERN USING THE SAME)”, Korea, **Application No.: 10-2024-0140399 (2024.10.15)**, Priority Application No.: 10-2023-0139441 (2023.10.18)
- 4) 조힘찬, 이재환, 연성범, “광민감성 무기 리간드 및 나노입자를 활용한 EUV 용 포토레지스트 조성물 및 이를 이용한 패턴 형성 방법 (PHOTORESIST COMPOSITION USING PHOTO-SENSITIVE INORGANIC LIGAND AND INORGANIC NANO-PARTICLES FOR EUV AND METHOD FOR FORMING PATTERN USING THE SAME)”, Korea, **Application No.: 10-2024-0138998 (2024.10.11)**, Priority Application No.: 10-2023-0159222 (2023.11.16)

- 5) 조힘찬, 이재환, 이승우, 김경민, 연성범, 이민구, 김근영, “반도체 소자, 이를 포함하는 이미지 처리 장치, 및 이미지 처리 장치의 동작 방법 (SEMICONDUCTOR DEVICE, IMAGE PROCESSING APPARATUS INCLUDING THE SAME, AND OPERATING METHOD OF IMAGE PROCESSING DEVICE)”, Korea,

Application No.: 10-2024-0101667 (2024.07.31)

- 6) 조힘찬, 하재영, 연성범, “유로퓸 기반 발광 나노결정 및 용액 합성을 통한 유로퓸 기반 발광 나노결정의 제조 방법 (EUROPIUM-BASED LIGHT-EMITTING NANOCRYSTALS AND METHOD FOR FABRICATING EUROPIUM-BASED LIGHT-EMITTING NANOCRYSTALS THROUGH COLLOIDAL SYNTHESIS)”, Korea,

Application No.: 10-2024-0041172 (2024.03.26)